Who Becomes Fuel Vulnerable, and When?

Early Warning of Household Vulnerability

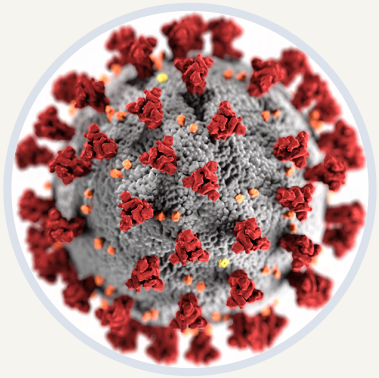
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WHY NOW?

The 2021–2024 energy crisis



Post-COVID
demand rebound



Russia–Ukraine
war, gas shock



Energy prices
rise 2021–2024



Households
in hardship

Prices rise in markets months before bills, yet help arrives only after hardship shows.

HOW HARDSHIP IS MEASURED TODAY

Too little, too late

Today's measures look backwards: they count hardship only **after** the bills are paid.

1 The 10% rule

Fuel spending of 10% or more of income, worked out from bills already paid.

Boardman (1991)

2 Low Income, High Costs

Low income plus high energy costs, from survey data a year or more old.

Hills (2012); now LILEE in England

3 Self-reported hardship

People say they could not keep warm or fell behind on bills, after it happened.

Household surveys

Our early warning works here

Today's measures only see it here: too late

Prices rise in markets → Bills arrive → Households struggle → Surveys record it → **Statistics published, help targeted**

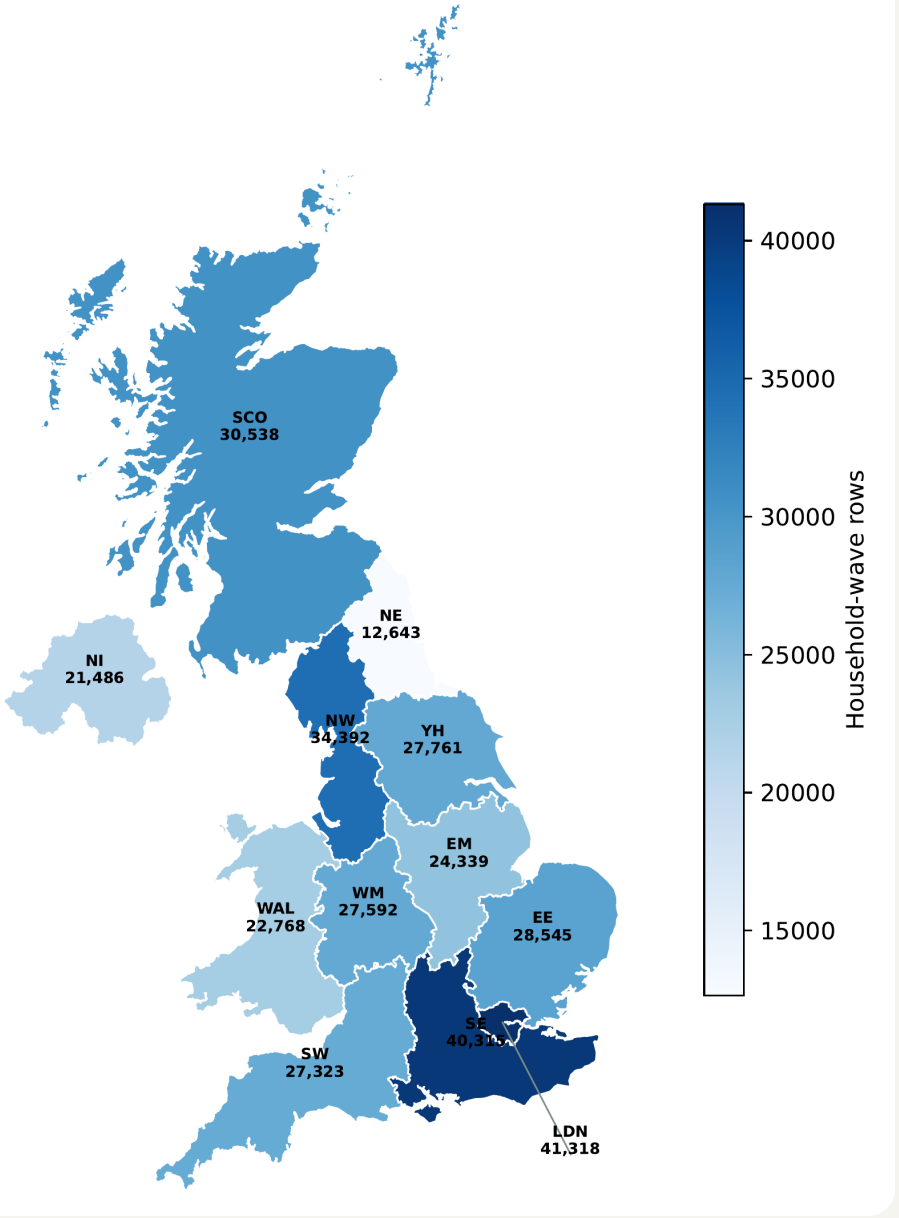
THE QUESTION

Can we tell, a year ahead, which households will cross the 10% line?

Who is exposed, and who can cope?

Where does risk differ from income poverty?

When can price forecasts warn us?



DATA

Sixteen years of UK households

339,201
household interviews (UKHLS)

12
UK regions

15
survey waves, 2009–2025

3
prices: gas, electricity, carbon

≥10% of net income spent on fuel = **at risk**

METHOD

How the early warning works

1 Collect

households, prices, JRF poverty

JRF

→

2 AI forecasts prices

SARIMA · Prophet · LSTM · TFT

→

3 Link to each home

forecast known on its interview date

→

4 Predict a year ahead

learn on earlier waves, test on 2022–25

→

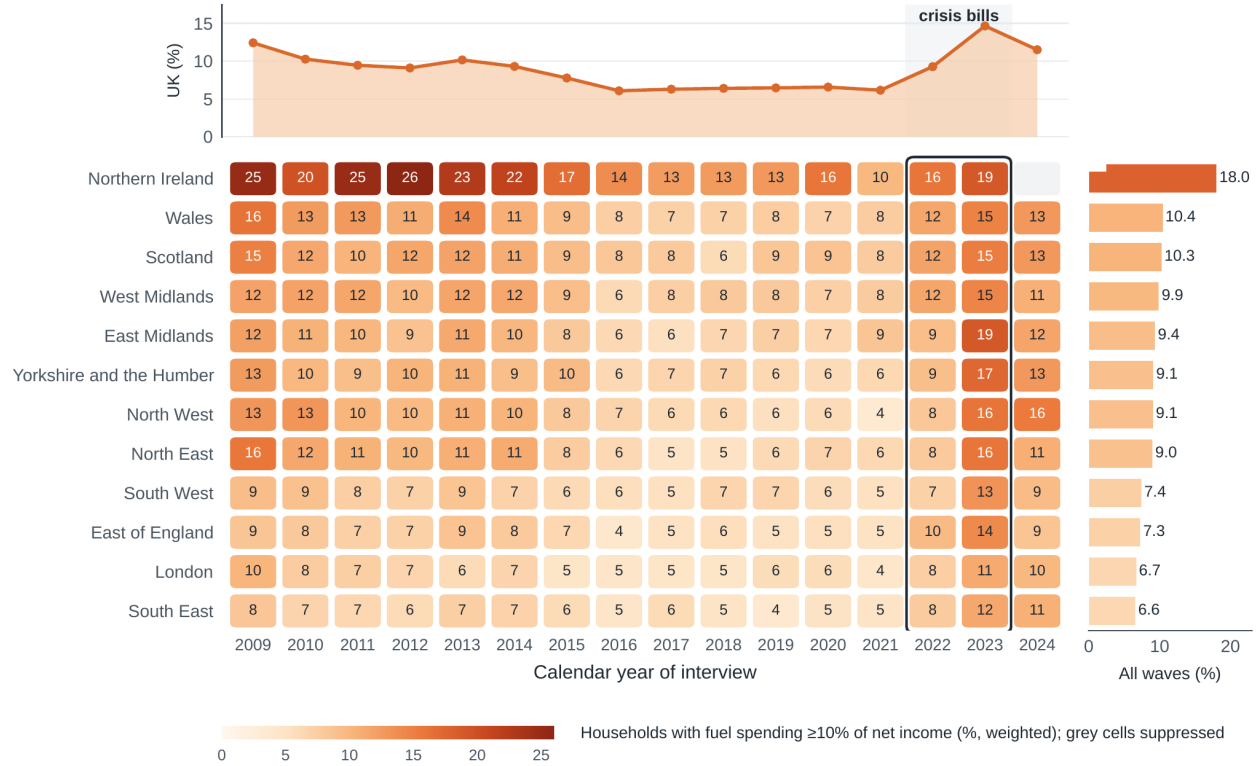
5 Check and compare

accuracy (AUC); rates vs JRF

[icon: bell] **Early-warning score for every household**

RESULT 1 · WHAT HAPPENED?

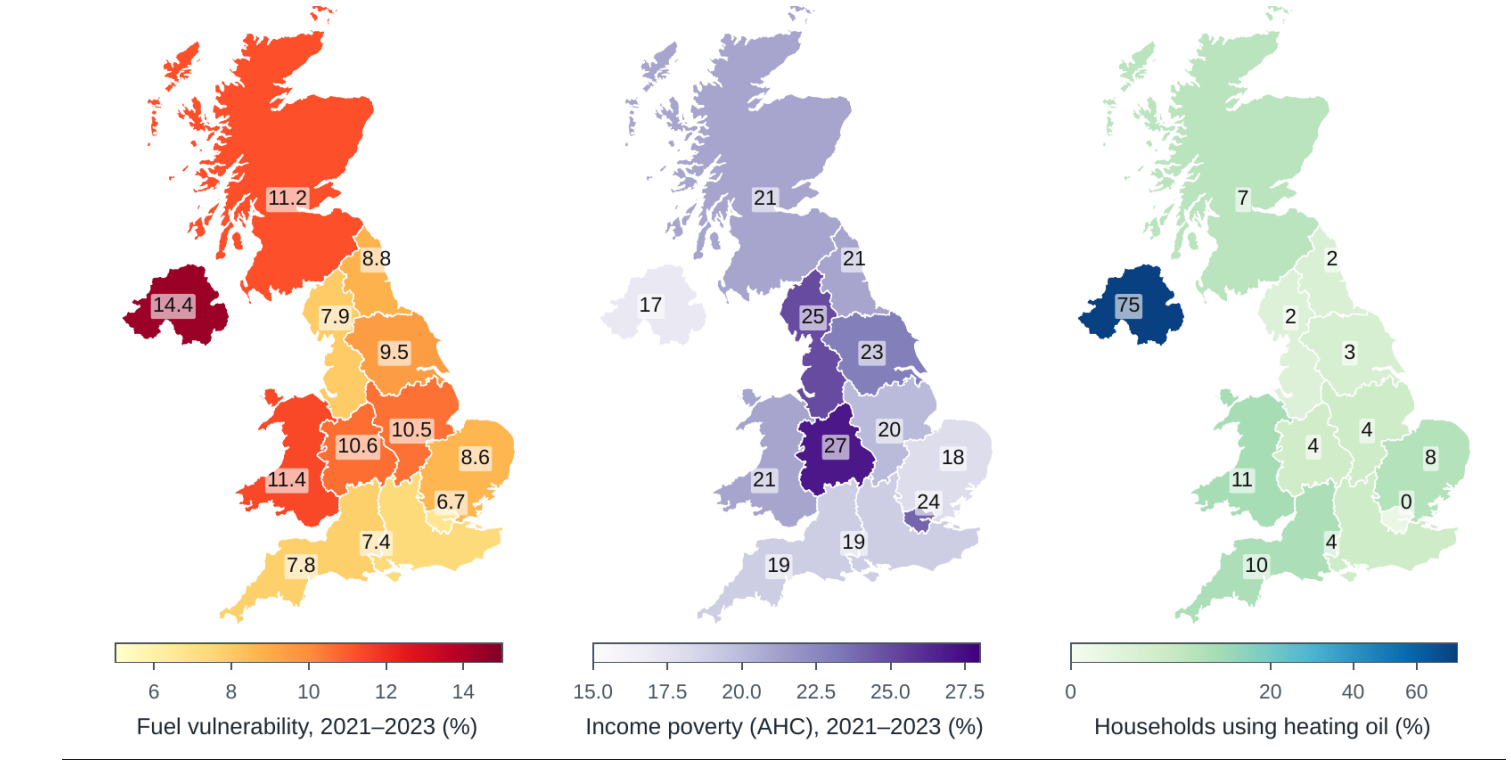
Risk more than doubled in the crisis



From about **6% (2021) to 15% (2023)**; **10 of 12** regions peaked in 2023.

RESULT 2 · WHAT HAPPENED?

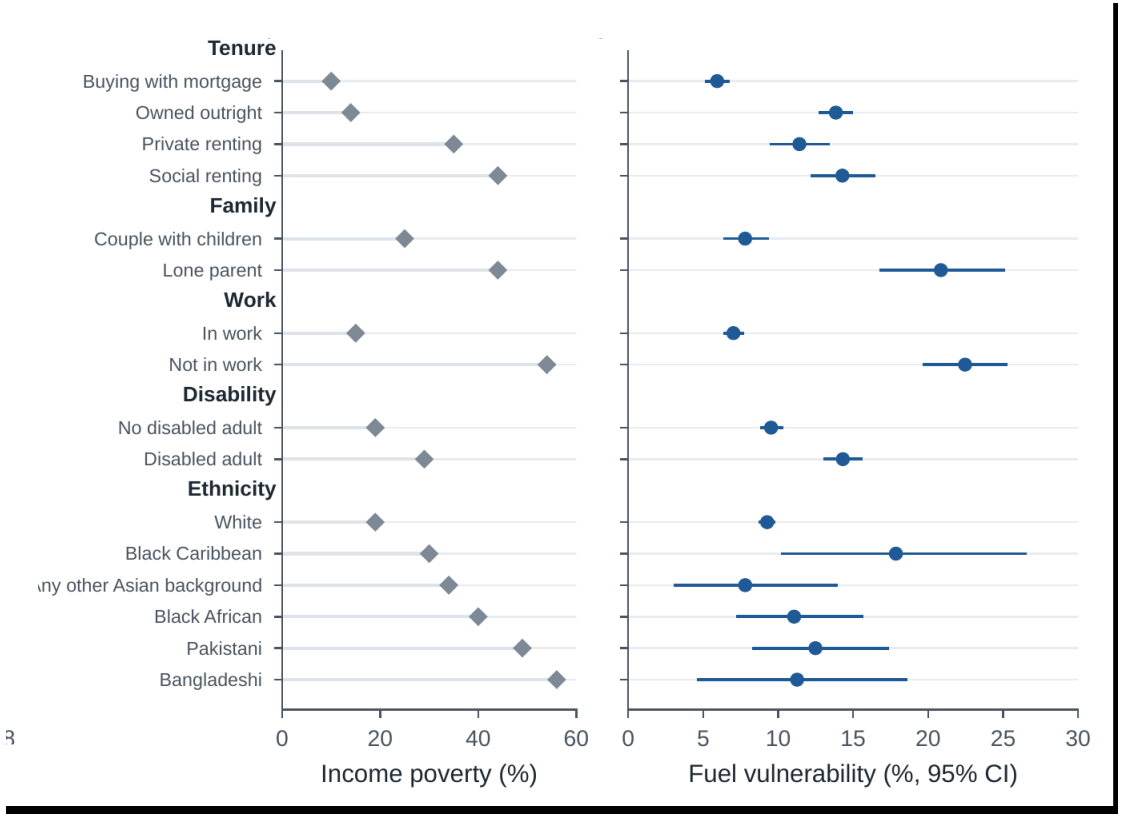
Fuel risk is not where poverty is



Northern Ireland: most fuel risk, least poverty, and **75%** of homes heat with oil.

RESULT 3 · WHO IS AT RISK?

The same people as JRF poverty



JRF

◆ JRF income poverty

● Our fuel vulnerability

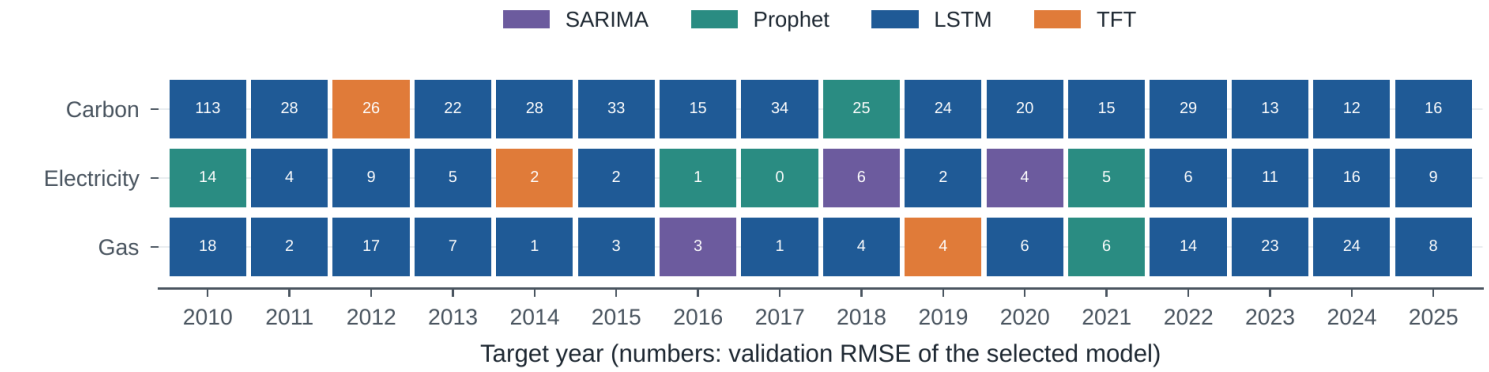
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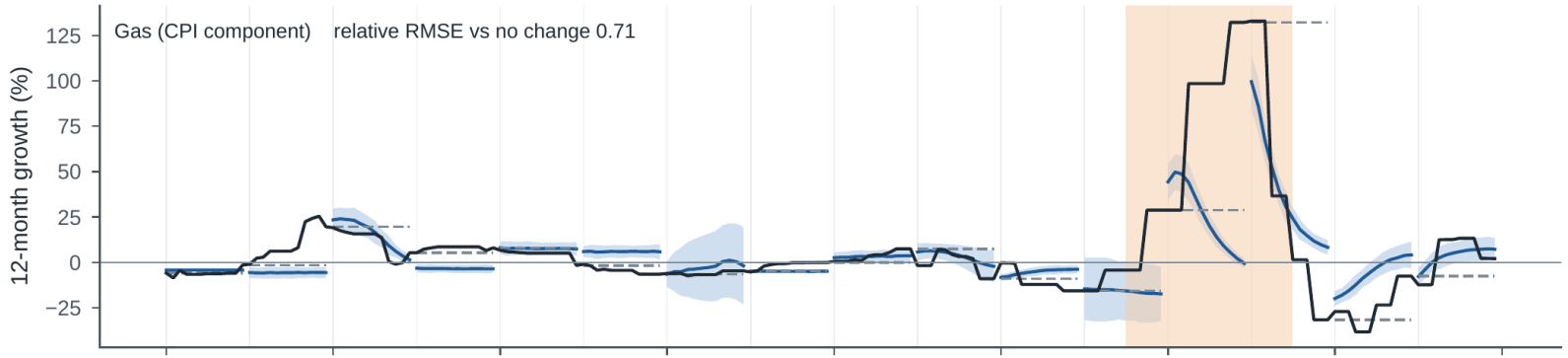
rank agreement by tenure

Renters, lone parents and those **not in work** are highest on both.

RESULT 4 · HOW?

AI forecasts energy prices

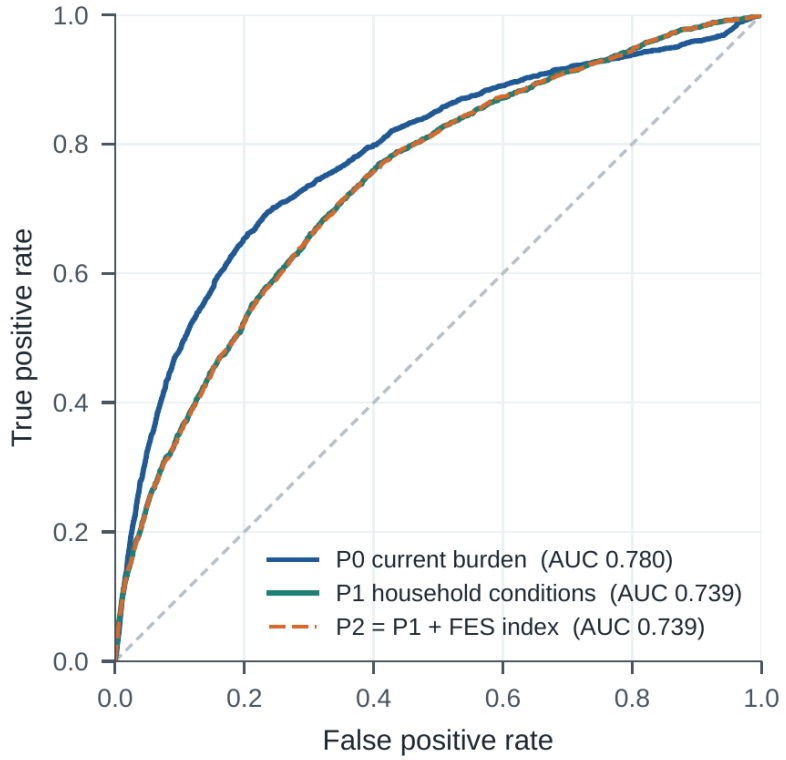


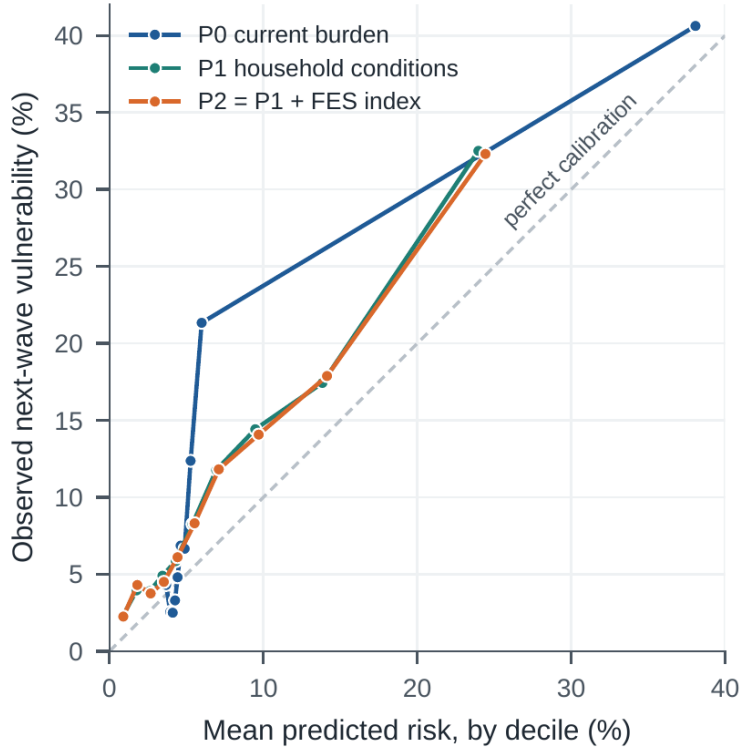


Deep-learning **LSTM** won **36 of 48** fuel-years; gas forecasts beat “no change” by **29%.**

RESULT 5 · CAN WE WARN?

A national forecast cannot say who

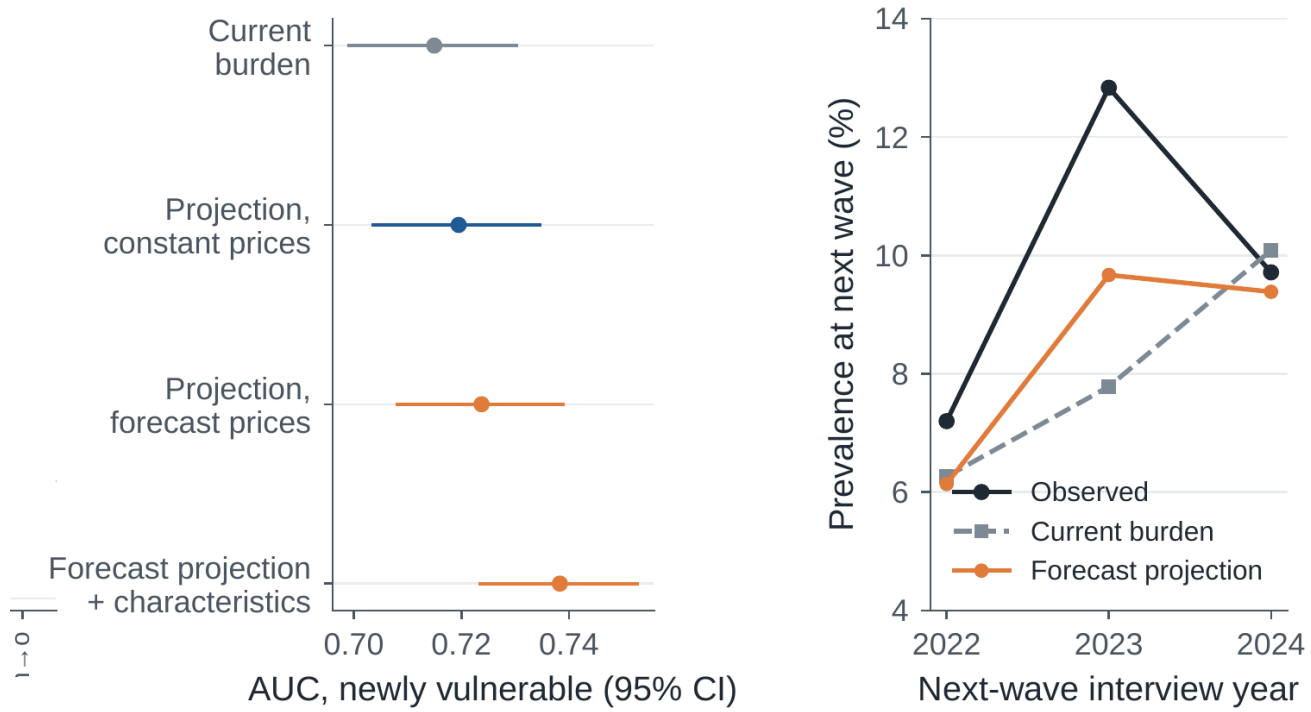




Today's bill warns best (**AUC 0.78**); the national index adds **nothing** (0.739 → 0.739).

RESULT 6 · CAN WE WARN? (EXPLORATORY)

Each home's own bill can



Forecast prices applied to each bill: best ranking (**AUC 0.738**) and closer to the 2023 peak.

WORKED EXAMPLE · SLIDE 11 VS SLIDE 12

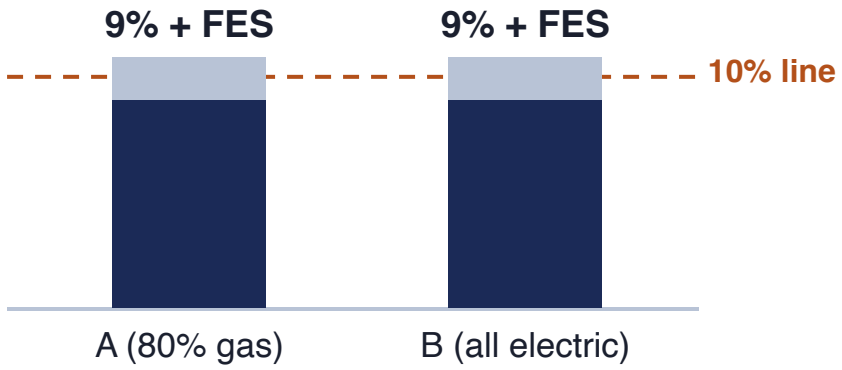
Same forecast, different homes

Forecast for next year: **gas +40%, electricity +10%.** Homes A and B both spend £1,800 of a £20,000 income on fuel today (9%). **A heats 80% with gas; B is all electric.**

Illustrative numbers.

✗ National forecast (slide 11)

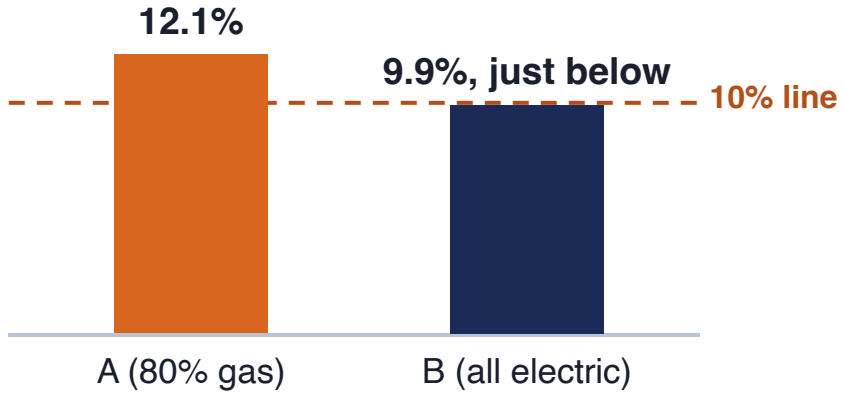
Both homes get the same national number



Ranked equal: it cannot say who

✓ Forecast projection (slide 12)

Each fuel's forecast applied to each home's bill



A is over the line, B is not: now we know who

Then add who can cope *(renting, lone parent, not in work)* → best ranking of all, **AUC 0.738**.

CONCLUSIONS

Forecasts say when. Each home's bill says who.

1. One price shock **doubled** fuel risk UK-wide.
2. Poverty finds the **people**, not the **places**.
3. A national forecast sizes support, but cannot target it.
4. Target support by **each home's bill**.

Thank you

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Code GitHub Scholar LinkedIn